Annie Prasanna Manoharan

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Education

University of Arkansas Aug 2024 – May 2026
Master of Science in Computer Engineering *Fayetteville, AR*
Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Privacy Enhancing Techniques

Sri Ramakrishna Engineering College Aug 2019 – May 2023
Bachelor of Engineering in Electronics and Communication *India*

Technical Skills

Cloud and Tools: REST API design, Distributed systems design, Design patterns, Time-series data modeling, Event-driven architecture, Financial market data, AWS Lambda, Agile/Scrum, Docker, Git, Linux, Grafana, Amazon S3, .NET, Data validation and quality, Multithreading, TDD, CI/CD, Amazon EC2
Data Engineering: Flink SQL, ETL/ELT, Apache Kafka, Apache Flink, ClickHouse, SQL Server, PostgreSQL, MySQL
Languages: SQL, C++, Python, C#
Machine Learning and AI:

Professional Experience

Data Engineering Intern Jul 2024 – Present
InsOps Inc. *Chicago, Illinois*

- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, and Apache Airflow for orchestration, with S3 cloud storage, for scalable data ingestion, transformation, validation, and analytics
- Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics

Teaching Assistant Sep 2024 – May 2026
University of Arkansas, Dept. of Computer Science and Engineering *Fayetteville, AR*

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals

Research Assistant Sep 2024 – Jan 2026
AICV Lab, University of Arkansas *Fayetteville, AR*

- Built reusable preprocessing workflows to segment, organize, and transform raw image/video data into structured inputs for downstream processing and evaluation
- Evaluated results using mAP, MOTA, IDF1, PSNR, SSIM, and FID

Project Engineer Jun 2023 – Jun 2024
Soliton Technologies *India*

- Developed customized software solutions for Dräger as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements
- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation

Projects

Time-Aware Market Data Analytics Service *C#, .NET, Apache Kafka, Python*

- Built a C# and .NET analytics service exposing REST APIs that serve real-time and historical queries over a time-aware data model with consistent point-in-time semantics
- Streamed tick-style event data through Apache Kafka topics into partitioned storage, generating and persisting derived series used by downstream research queries
- Wrote Python workers to compute derived analytics over the event stream, with unit tests, code review discipline, and monitoring hooks following SDLC best practices

Real-Time Medical IoT Analytics Platform *Apache Kafka, Apache Flink, Flink SQL, Python, ClickHouse, Grafana*

- Developed an end-to-end, sub-second latency streaming platform using Apache Flink SQL and Apache Kafka to ingest, process, and score high-frequency medical IoT device data
- Implemented temporal Flink OVER windows partitioned by device ID to compute rolling Z-scores over 5-second sliding intervals, applying custom watermark delay strategies to handle out-of-order streams